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Acceptance Test deliverable

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Chapter 1

Introduction and Project Identification

1.1 Purpose

This document reports the results of the Acceptance Testing phase performed on the **BestBikePaths (BBP)** application. The goal is to validate the software against the requirements specified in the RASD, verify the consistency of the architectural design, and assess the reproducibility of the installation process.

1.2 Reviwed Project Identification

The software project analyzed in this document is **BestBikePaths**. The application was designed and developed by the following team:

- **Andrea Fidanza** (Mat. 10730617)
- **Guglielmo Gatti** (Mat. 10794407)
- **Marco Maiocchi** (Mat. 10723893)

1.3 Source Code Repository

The testing procedures were performed using the source code, database dumps, and binary files hosted at the following public GitHub repository:

`https://github.com/topox19547/FidanzaGattiMaiocchi`

Specific focus was placed on the `/Implementation` folder for the source code and the `/DeliveryFolder` for the documentation and executable binaries.

1.4 Reference Documents

The evaluation was conducted based on the following official documents provided by the development team, which served as the "oracle" for expected system behavior:

1. **RASD** (Requirement Analysis and Specification Document) - Version 1.0
2. **DD** (Design Document) - Version 1.0
3. **ITD** (Implementation and Test Document) - Version 2.0

1.5 Testing Team

This Acceptance Test Document (ATD) has been produced by:

- **Alessio Pesavento**
- **Alberto Spezzapria**
- **Luca Suzani**

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Chapter 2

Installation Setup

This section describes the activities performed to install the *Best Bike Paths* (BBP) system prototype and analyzes the overall experience, highlighting both the effectiveness of the instructions provided in the ITD document and the technical issues and gaps encountered.

2.1 Installation Procedure

The installation was executed by faithfully following the instructions reported in Chapter 6 of the ITD document. The environment was configured on a single machine hosting both the server and the client (via emulator), as suggested by the prerequisites. The procedure followed the indicated logical order:

1. Configuration of the PostgreSQL database and the PostGIS extension.
2. Import of the schema and data via the `DatabaseDump.sql` file.
3. Startup of the Server component using the provided `.jar` executable.
4. Installation of the Android Client via drag-and-drop of the `.apk` file into the emulator.

2.2 Documentation Strengths

The analysis of the installation guide highlighted several positive aspects that facilitated the initial startup:

- **Availability of Executables:** The provision of pre-compiled files `BestBikePathsServer.jar` and `BestBikePathsClient.apk` was fundamental. This allowed testing the system without necessarily having to recompile the entire source code, reducing potential build errors in the first instance.
- **Logical Sequence:** The structure of Chapter 6 is well-organized, guiding the user through the application layers in the correct order (Database $\rightarrow$ Server $\rightarrow$ Client), which is essential to avoid connection errors during the first startup.
- **Explicit Commands:** The commands provided for importing the database and starting the server (including flags to overwrite the default username and password) proved to be syntactically correct and functional.

2.3 Critical Issues and Problems Encountered

Despite the logical structure, the installation phase proved to be significantly time-consuming and complex due to various incompatibilities and informational gaps:

- **Software Compatibility Issues:** The use of different or older versions created conflicts that required time to resolve, forcing a non-trivial update or specific configuration of the environment.
- **Missing Configuration Details:** Although the main commands are present, the guide tends to take prior knowledge of specific tools for granted, creating difficulties for users not expert with certain software:
  - **Database Management:** The instructions for importing the SQL dump (`\i 'path/to/file'`) lack details regarding potential file read permission issues or path handling on different operating systems, which often generate silent errors.
  - **External API Configuration:** For those wishing to compile the client, the instructions on API key management are too brief. It is indicated to insert them into the `local.properties` file, but no guide is provided on how to generate these keys or which specific SDKs to enable in the respective developer consoles, risking having an installed app with non-functional maps.
- **Overall Setup Time:** Due to the aforementioned issues, the actual time required to obtain a functioning system was much higher than expected, turning what should have been a simple prototype installation into an execution environment debugging process.

In conclusion, the installation process proved to be **notably long and burdensome**. Although the documentation covers the necessary logical steps, the practical experience highlighted several points of friction due to software incompatibilities and a level of detail not always adequate for those approaching the ecosystem from scratch.

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Chapter 3

Acceptance Test Cases and Results

Chapter 4

Documentation and Code Quality

This section provides an evaluation of the delivered documentation. It analyzes the consistency between the initial specifications, the architectural choices and the final software product, highlighting both the theoretical work and the practical gaps encountered during the testing phase.

4.1 RASD (Requirement Analysis and Specification Document)

Strengths

- **Formalization and Safety Logic (Alloy):** The inclusion of a complete Alloy model is a point of excellence. Specifically, the **SafetyFirst** assertion (Image 4.2) formally defines the ranking logic, ensuring that dangerous paths are pessimistically penalized. This demonstrates a strong commitment to verifying the domain consistency before implementation.
- **Comprehensive Modeling:** The document provides extensive UML modeling, including Class Diagrams and State Diagrams, which effectively clarify the domain entities and their lifecycle.
- **Detailed Scenarios:** The description of usage scenarios (e.g., *Scenario 3: Manual Tracking*) is narrative and effective, helping the reader understand the expected logical flow "on paper."

Weaknesses

- **Missing Non-Functional Constraints (UX):** The document fails to specify critical usability constraints. The **150-meter limit** for point creation—which acts as a blocking factor in the software—is not mentioned in the specific requirements nor in the system limitations. This creates a discrepancy between the expectation created by the RASD (freedom of tracking) and the reality of the software.
- **Ambiguity of "Crowd-sourcing":** The RASD places emphasis on the social nature of the app. However, it does not address the risk of user base fragmentation resulting from the choice of a single platform (Android). This decision is taken for granted without an impact analysis on the core requirement of "massive data collection."

4.2 DD (Design Document)

Strengths

- **Architectural Layering:** The high-level architecture is clearly defined. The document effectively visualizes the separation of concerns through a 3-tier structure (Client, Server, Database), and the decomposition into modules (e.g., *Path Scoring*, *Trip Management*) is coherent with the functional requirements.

- **Integration Strategy:** The document outlines a progressive integration plan (described in steps, e.g., Step 9 to Step 11), showing a logical approach to assembling the backend and frontend components.

Weaknesses

- **Unjustified Over-Engineering:** The choice of the technology stack (Spring Boot + PostgreSQL/PostGIS) is presented as a given, without a substantial *Rationale*. For a system managing simple geospatial entities, the absence of a discussion on lighter alternatives is an architectural oversight, leading to high development friction.
- **UI/UX Discrepancy:** The mockups and interface descriptions in the DD do not anticipate the complexity of map management (e.g., obscured play buttons, involuntary point deletion), suggesting an interface design that underwent little iteration.

4.3 ITD (Implementation and Test Document)

Strengths

- **Intellectual Honesty:** Section *2.3 Excluded Functionalities* clearly lists what has not been implemented, allowing the reviewer to avoid searching for non-existent features.
- **Adherence to Code Standards:** The described source code structure (MVC pattern on Spring, standard Android structure) appears to follow the best practices of the respective frameworks, despite the complexity.

Weaknesses

- **Non-Standardized Deployment:** In a modern DevOps context, requiring manual DB installation, specific user creation, and manual SQL script injection is considered an anti-pattern. The absence of a `docker-compose` file forces the reviewer to handle compatibility issues (PostGIS, Java versions) manually. Details on `Installation Setup` section.
- **Testing Friction:** No data seeding scripts were provided. Delivering an empty database for an app that derives its value from data (paths, ranking) prevents immediate testing and forces the reviewer to manually handle data entry.

4.4 Presentation and Editorial Quality

Although they do not directly impact software functionality, we detected some graphical and editorial inconsistencies that reduce the perceived quality and navigability of the documentats.

- **Lack of Uniformity:** There is no common visual identity among the documents. The RASD was evidently generated via Notion export, resulting in a style disconnected from the DD and ITD, which were drafted in LaTeX. This fragmentation leads to the absence of a common header/footer and inconsistent formatting styles, giving the impression of "disjointed" documents rather than a unified project corpus.
- **Absence of Dynamic Navigation:** Despite the use of LaTeXfor the DD and ITD, the generated PDFs are "static." Essential features for digital consultation, such as the bookmarks sidebar and clickable internal cross-references, are missing. In technical documents spanning dozens of pages, the lack of these tools makes navigation slow and cumbersome.

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- **Attention to Detail (Typos & Formatting):** Several minor editorial inaccuracies were found, such as inconsistent use of bold text to highlight key terms, capitalization errors, and non-uniform caption formatting. While minor, these details contribute to a less polished and professional overall appearance.

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Chapter 5

Effort Spent

Task	Date	Alessio P.	Alberto S.	Luca S.
Installation & Testing	04/02/2026	1h	1h	1h
	05/02/2026	2.5h	2h	1.5h
	06/02/2026	1h	1h	1.5h
	Subtotal	**4.5h**	**4h**	**4h**
Document Preparation	06/02/2026	1h	1h	1h
	08/02/2026	2h	3h	2h
	Subtotal	**3h**	**4h**	**3h**
TOTAL EFFORT		**7.5h**	**8h**	**7h**

Table 5.1: Hours of work spent by each author on installation, acceptance testing and document preparation